3789. Minimum Cost to Acquire Required Items

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Problem and Solution

We're given $cost1, cost2$, and $costBoth$, which are the costs of purchasing 1 unit of type 1, 1 unit of type 2, or 1 unit of both type 1 and type 2, respectively. The requirements per type are also given, $need1$ and $need2$, representing the minimum quantity needed for the respective type. The problem asks: "What is the minimum possible cost to satisfy both of these requirements?"

To begin, it helps to think simple and consider some cases: What if equal quantities of type 1 and type 2 are needed? (i.e. $need1 = need2$) Intuitively, we have two ways to obtain type 1 and type 2:

- (1) Purchase both types individually, paying both $cost1$ and $cost2$.
- (2) Purchase both types together, paying $costBoth$.

How do we decide which method to choose? Whichever is cheaper, or course. Because we need to buy the same amount for both types, $n = need1 = need2$, we'll have to pay either $n * (cost1 + cost2)$ or $n * costBoth$. The cheaper option is the solution!

The next case to consider is when only one type is needed, consider this type to be type i. There are two methods to obtain type i: purchasing it individually and paying $costi$ per unit, or purchasing both types and paying $costBoth$ per unit. Of course, when purchasing both we do indeed end up with additional units of the other type. However, this is explicitly stated within the problem as permissible, so long as we acquire the minimum quantities of each type.

As with our first examined case, we end up with two possible options for purchasing $needi$ units of type i. Likewise, the solution is similarly trivial: Take the cheapest option. Purchasing just type i will cost $needi * costi$, and purchasing both types to obtain type i will cost $needi * costBoth$. Whichever cost is the lowest, that's what we'll take!

Taking a step back, its evident that these two cases can be used to solve every case of input. Obviously, if our input falls into one of these cases, then we already know the solution. If not, we know that $need1 \neq need2$ and $need1, need2 > 0$. Given this, we know we need some common quantity of

type 1 and type 2, up to a certain point. Whichever the smaller requirement, call it *needi*, we need *needi* units of both type 1 and type 2. This can be handled with our first case, considering the common quantity required by both types.

Following this, we know that there's only one more requirement to satisfy, because we already satisfied the smaller requirement *needi*. This can be handled with our second case, considering the remaining quantity required by the larger type.

During these previous steps, all that's needed is to keep track of the total amount spent on purchasing units of each type.

Algorithm

- 1. Purchase equally as much of both types as possible, without exceeding either requirement. (individually or together, whichever costs less)
- 2. Purchase the remaining quantity needed of the appropriate type, if any, to satisfy the larger requirement. (individual or combined with the other type, shichever costs less)
- 3. Return the total amount spent purchasing units

Analysis of Running Time

The solution requires only constant-time operations and conditionals, across all input. This solution runs in constant-time, $O(1)$.